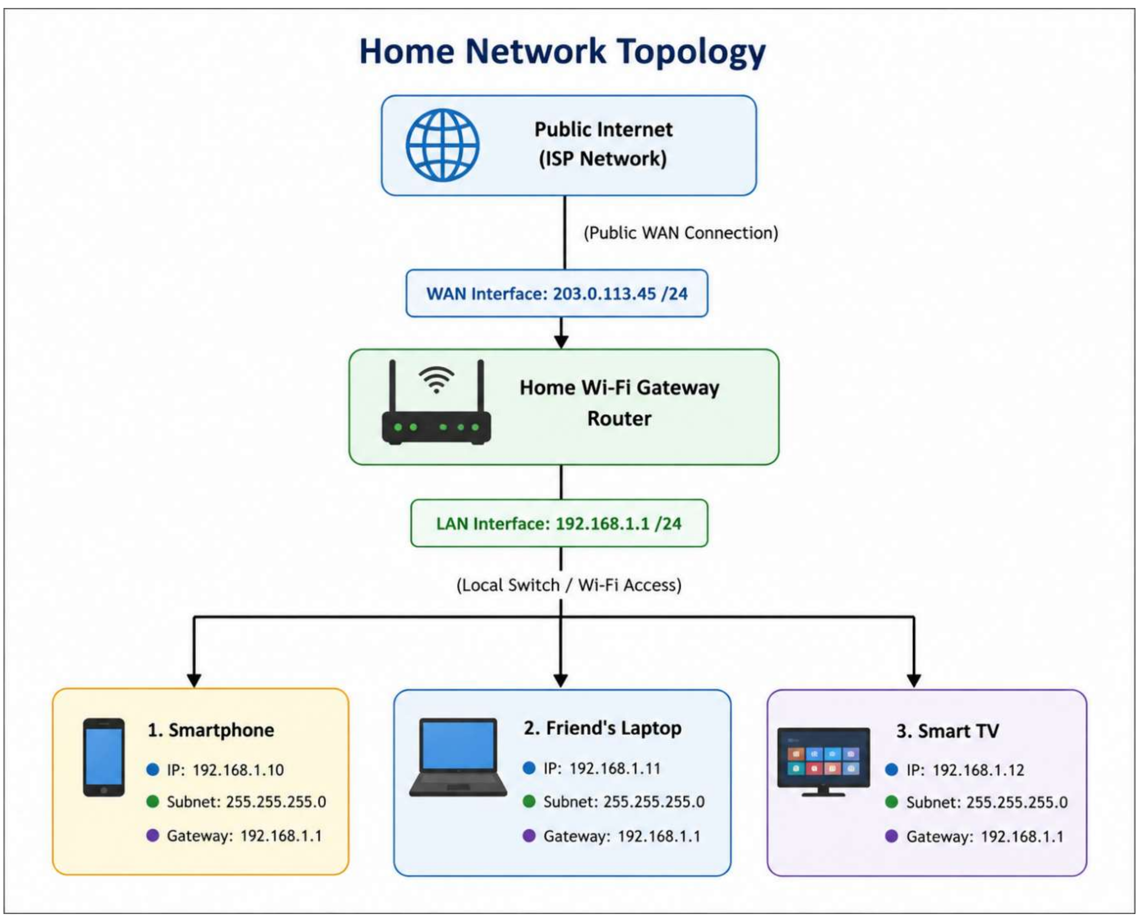
Task 1:

Home Network Topology Diagram & IP Address Assignment

Objective: Diagram a small local network showing three end-user devices connected through a central router, explicitly labeling private LAN IP addresses and public/private router interfaces.

1. Network Architecture Diagram:



2. Device & Interface Addressing Table:

Network Component	**Interface / Connection**	**Assigned IP Address**	**Subnet Mask**	**Default Gateway**
Router (WAN)	External (ISP Side)	203.0.113.45	255.255.255.0	203.0.113.1 (ISP Hop)
Router (LAN)	Internal Gateway	192.168.1.1	255.255.255.0	N/A (Self Gateway)
Smartphone	Wi-Fi Interface	192.168.1.10	255.255.255.0	192.168.1.1
Friend's Laptop	Wi-Fi Interface	192.168.1.11	255.255.255.0	192.168.1.1

Network Component	Interface / Connection	Assigned IP Address	Subnet Mask	Default Gateway
Smart TV	Wired Ethernet	192.168.1.12	255.255.255.0	192.168.1.1

Task 2:

Manual IP Routing Simulation Across Subnets

Objective: Simulate step-by-step packet propagation from a local host to a remote host on a different subnet, detailing routing table lookup decisions at each intermediate hop.

1. **Network Setup Details:**

- **Source Host (Your Laptop):** 192.168.1.50/24 | Default Gateway: 192.168.1.1
- **Destination Host (Friend's PC):** 10.0.2.100/24 | Default Gateway: 10.0.2.1
- **Router 1 (Local Gateway):**
  - LAN Interface (eth0): 192.168.1.1/24
  - WAN Link (eth1): 172.16.0.1/30
- **Router 2 (Remote Gateway):**
  - WAN Link (eth0): 172.16.0.2/30
  - LAN Interface (eth1): 10.0.2.1/24

---

2. **Step-by-Step Packet Traversal & Routing Decisions:**

Step 1: Source Host (Your Laptop) Initial Decision

- The laptop compares the destination IP (10.0.2.100) against its own subnet (192.168.1.0/24).
- Since the destination is on a **different subnet**, the laptop encapsulates the packet and forwards it to its Default Gateway (**Router 1** at 192.168.1.1).

Step 2: Hop 1 - Router 1 (Local Router) Decision

- Router 1 receives the packet on interface eth0 and inspects the Routing Table for 10.0.2.100:

Destination Network	Subnet Mask	Next Hop (Gateway)	Outgoing Interface
192.168.1.0	255.255.255.0	Directly Connected	eth0
172.16.0.0	255.255.255.252	Directly Connected	eth1

Destination Network	Subnet Mask	Next Hop (Gateway)	Outgoing Interface
10.0.2.0	**255.255.255.0**	**172.16.0.2**	**eth1**

- **Decision:** Match found for network 10.0.2.0/24. Router 1 forwards the packet out of interface **eth1** toward Next Hop **172.16.0.2** (Router 2).

Step 3: Hop 2 - Router 2 (Remote Router) Decision

- Router 2 receives the packet on interface eth0 and inspects its Routing Table:

Destination Network	Subnet Mask	Next Hop (Gateway)	Outgoing Interface
172.16.0.0	255.255.255.252	Directly Connected	eth0
10.0.2.0	**255.255.255.0**	**Directly Connected**	**eth1**

- **Decision:** The destination subnet 10.0.2.0/24 is directly connected to interface **eth1**.
- Router 2 resolves the MAC address of 10.0.2.100 via ARP and delivers the packet directly to your friend's laptop.

Task 3:

Routing Table for Zomato Multi-City Branch Network

Objective: Design a central router routing table for a multi-city food delivery organization (Zomato) connecting three distinct branch subnets (Mumbai, Delhi, Bengaluru) and a default internet route.

---

1. Network Subnet Architecture:

- **Central Router:** Mumbai HQ Main Core Router
- **Branch 1 (Mumbai HQ):** 10.10.0.0/24 (Directly Connected)
- **Branch 2 (Delhi Regional Office):** 10.20.0.0/24 (Connected via Serial/WAN link)
- **Branch 3 (Bengaluru Logistics Center):** 10.30.0.0/24 (Connected via Serial/WAN link)
- **Default Internet Route:** 0.0.0.0/0 (Public ISP Gateway)

---

2. Complete Master Routing Table:

Destination Branch / Network	Network Address	Subnet Mask	Next Hop IP Address	Outgoing Interface	Metric / Route Type
Mumbai Branch (Local LAN)	10.10.0.0	255.255.255.0	Directly Connected	GigabitEthernet0/0	0 (Connected)
Delhi Branch Network	10.20.0.0	255.255.255.0	192.168.12.2	GigabitEthernet0/1	1 (Static / OSPF)
Bengaluru Hub Network	10.30.0.0	255.255.255.0	192.168.13.2	GigabitEthernet0/2	1 (Static / OSPF)

Destination Branch / Network	Network Address	Subnet Mask	Next Hop IP Address	Outgoing Interface	Metric / Route Type
Default Internet Gateway	0.0.0.0	0.0.0.0	203.0.113.1	WAN0 (Serial 0/0/0)	1 (Default Static Route)

Task 4:

Static Routing vs. Dynamic Routing Protocol Comparison

Objective: Compare Static Routing against Dynamic Routing protocols (e.g., OSPF, RIP) detailing 3 pros and 3 cons of each, complemented by real-world application use cases (e.g., Zomato food delivery app backend infrastructure).

---

1. Comparison Matrix: Static vs Dynamic Routing

Static Routing:

- **Pros:**
  1. **Zero CPU/RAM Overhead:** Does not require routing protocol updates or background calculation algorithms.
  2. **High Security & Predictability:** Network administrator maintains complete control over traffic paths; no unauthorized route advertisements.
  3. **No Network Bandwidth Consumption:** Zero routing protocol packets (hello timers, link-state updates) are transmitted over links.
- **Cons:**

1. **Lack of Scalability:** Highly complex and prone to human error when managing large or growing enterprise networks.
2. **No Automatic Failover:** If a primary link fails, traffic is dropped until an administrator manually changes the route.
3. **High Maintenance Cost:** Every network modification requires manual recalculation and reconfiguration across all routers.

Dynamic Routing (e.g., OSPF, BGP, RIP):

- **Pros:**
  1. **Automatic Convergence & Failover:** Instantly reroutes traffic around link failures or network congestion without manual human intervention.
  2. **High Scalability:** Effortlessly scales to large, complex infrastructure networks across multiple geographies.
  3. **Load Balancing & Optimal Pathing:** Dynamically selects the fastest path based on real-time network metrics (bandwidth, latency, cost).
- **Cons:**

1. **High Resource Consumption:** Uses CPU cycles, RAM, and bandwidth to continuously send hello packets and process routing algorithms (e.g., Dijkstra's algorithm in OSPF).
2. **Security Vulnerabilities:** Susceptible to routing table poisoning or rogue router advertisements if dynamic authentication is not configured.
3. **Configuration & Operational Complexity:** Requires specialized knowledge to design, troubleshoot, and optimize protocol timers.

---

2. Real-World Application Example (Food Delivery App - Zomato):

- **Why Zomato Benefits from Dynamic Routing:**
  - **Zero Downtime Food Ordering:** When millions of users place orders simultaneously, backend servers communicate across microservices distributed in different cloud regions.
  - **Automatic Failover during Traffic Spikes:** If a primary data link connecting Zomato's Order

Processing Service to Payment Gateways suffers a fiber cut, dynamic routing protocols (like OSPF/BGP) detect the failure within milliseconds and automatically switch traffic over to redundant secondary links.

- **Static Routing Usage:** Zomato might still use static routes for non-critical, fixed internal administrative setups like linking a point-to-point backup printer server in a local branch office.

Task 5:

OSPF Routing Configuration & Line-by-Line Annotation

Objective: Document a sample OSPF routing configuration between two Cisco routers and annotate each CLI command line to explain its technical functionality.

. AI-Generated Cisco IOS OSPF Configuration:

Router 1 (R1) Configuration:

R1> enable

R1# configure terminal

R1(config)# router ospf 1

R1(config-router)# router-id 1.1.1.1

R1(config-router)# network 192.168.1.0 0.0.0.255 area 0

R1(config-router)# network 10.0.0.0 0.0.0.3 area 0

R1(config-router)# exit

Router 2 (R2) Configuration:

R2> enable

R2# configure terminal

R2(config)# router ospf 1

R2(config-router)# router-id 2.2.2.2

R2(config-router)# network 192.168.2.0 0.0.0.255 area 0

R2(config-router)# network 10.0.0.0 0.0.0.3 area 0

R2(config-router)# exit

2. Line-by-Line Explanation & Annotation:

- **enable**: Enters privileged EXEC mode from user mode, granting access to administrative CLI commands.
- **configure terminal**: Enters global configuration mode to make system-wide changes to the router.
- **router ospf 1**: Enables the OSPF dynamic routing process on the router using **Process ID 1** (a local identifier used internally by the router).
- **router-id 1.1.1.1**: Manually assigns a unique 32-bit 1.1.1.1 identifier to R1 (and 2.2.2.2 to R2) so other routers can identify it during neighbor adjacency formation.
- **network 192.168.1.0 0.0.0.255 area 0**:

    - Tells OSPF to activate the protocol on any interface matching the 192.168.1.0 IP subnet.
    - Uses wildcard mask 0.0.0.255 (inverse of 255.255.255.0).
    - Assigns this subnet to **Backbone Area 0** (area 0).
- **network 10.0.0.0 0.0.0.3 area 0**: Activates OSPF on the point-to-point inter-router link network (10.0.0.0/30) in Area 0 so R1 and R2 can exchange OSPF Hello packets and form dynamic neighbor relationships.
- **exit**: Exits router configuration mode and returns to global configuration mode.

---

3. Verification Command:

After applying the configuration, execute the following command on R1 to verify OSPF neighbor adjacency with R2:

R1# show ip ospf neighbor

Expected Status: The neighbor state should show **FULL/DR** or **FULL/BDR**, indicating complete Link-State Database (LSDB) synchronization between the two routers.